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DNA profile match probability calculation: how to allow for population stratification, relatedness, database selection and single bands

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Abstract

In DNA profile analysis, uncertainty arises due to a number of factors such as sampling error, single bands and correlations within and between loci. One of the most important of these factors is kinship: criminal and innocent suspect may share one or more bands through identity by descent from a common ancestor. Ignoring this uncertainty is consistently unfair to innocent suspects. The effect is usually small, but may be important in some cases. The report of the US National Research Committee proposed a complicated, ad-hoc and overly-conservative method of dealing with some of these problems. We propose an alternative approach which addresses directly the effect of kinship. Whilst remaining conservative, it is simple, logically coherent and makes efficient use of the data.

Key words: DNA profiles; Population genetics; Evidence; Statistics; Databases

1. Statistical issues

1.1. Problems with the product rule

Most current analyses of a match of two DNA profiles in forensic identification involve using the so-called 'product rule' to obtain an estimate of the profile match probability in terms of relative frequencies of alleles in a database. If the calculation ignores uncertainty then the resulting value overstates the strength of the evidence.

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In typical cases the evidential impact of a match is very strong and the overstatement of the strength of evidence by perhaps one or two orders of magnitude may have little practical effect. However, the effect can be important in some cases involving, for example, migrant groups with few representatives in the database, closely related individuals among the possible suspects or incomplete profiles. Even when the match probability is small, if there is little or no other evidence then one or two orders of magnitude may substantially affect the outcome.

We will refer to the use of the product rule which ignores uncertainty as 'simple'. Aspects of the simple product rule have been criticized by a number of authors [1–5]. Empirical evidence has recently been provided by Krane et al. [6], who highlighted variation between databases and higher match probabilities in sub-populations than in large populations. They calculated match probabilities within a group of Italians and a group of Finns and found that values obtained using a mixed Caucasian database were typically much lower than those obtained using the appropriate databases. These authors also reported two three-locus matches among 70 individuals in the Italian database; the matches might be between close relatives. In forensic databases constructed from casework, it is difficult to draw firm inferences about match probabilities from observations of whole-profile matches, since they are very rare and an observed match may be taken as proof that the two profiles are from the same individual.

The recent report of the US National Research Council (NRC) [7] identified some of the problems with the product rule. It proposed modifying the rule by replacing racial group sample frequencies with values obtained by a combination of the 'ceiling principle' (using the largest value from a range of databases) and upper confidence limits, subject to an overall minimum of 10% as an interim measure, which may be reduced to 5% after further research. Although the report explains clearly the underlying statistical and genetic issues, its recommendations for overcoming the problems have been widely criticized as complicated and ad-hoc [8]. The ceiling principle is unsatisfactory in taking into account populations which have no possible connection with a given crime. In applying the principle, it is difficult to decide how many sub-populations to consider and which levels of stratification should be investigated. The NRC report gives little attention to important statistical issues and its statistical analysis contains errors [9–10].

We propose an alternative modification to the product rule which is logically coherent and conservative while, we believe, not excessively understating the high discriminatory power of the DNA profile method. Our approach uses established statistical and genetical principles to make appropriate allowances for uncertainty. It addresses the problem of non-independence due to shared ancestry, and clarifies the issues by specifying a readily-interpreted parameter which measures the amount of shared ancestry. The work reported here develops that published previously [4–5].

Several authors have published work which tends to show that any error following from the use of the simple product rule is usually small [12–16]. This work has served to diminish some concerns about the effects of population structure such as Hardy-Weinberg and linkage disequilibrium. However, we now briefly outline two fundamental problems which have not been addressed.

1.2. Uncertainty in identification evidence

Ignoring uncertainty in the assessment of forensic identification evidence leads to an overstatement of the weight of the evidence. Suppose that the relative frequency of a DNA profile in a particular population is known to be p. If two individuals are drawn randomly from the population then the probability that the second has the profile given that the first has it is simply

$$P[\text{Match}] = p \tag{1}$$

In practice, the profile relative frequency is not known exactly. If σ^2 denotes the variance of the profile relative frequency about its expected value p, then the match probability is

$$P[\text{Match}] = p + \sigma^2/p \tag{2}$$

The error due to uncertainty about p does not cancel out. The reason is that under the hypothesis of innocence we have observed two copies of the profile, which makes it likely that the profile is more common in the population than we had previously believed. For fixed σ, the effect becomes greater as p becomes small. For a further discussion, see Ref. 18.

Biologists will be familiar with a particular case of Eq. (2) known as the Wahlund effect (see for example Ref. 17). It concerns the probability that an individual is homozygous and hence has two matching alleles. In a subdivided population there can be more homozygotes than would be expected from the average allele frequency. This is a consequence of the variation in allele frequency amongst the sub-populations.

Uncertainty about the relative frequency p of a DNA profile is due to several factors, both statistical and genetic. Typically values for σ^2 in Eq. (2) are small, but since p is often very small, the effect of this uncertainty can be relatively large. Database allele frequency estimates are subject to some uncertainty due to sampling variation and the fact that an apparent homozygote may correspond to one or two observations of an allele. There is also uncertainty due to doubts about the validity of the assumptions of Hardy-Weinberg equilibrium and independence across loci. These effects are widely believed to be small. More importantly, criminal and innocent suspects may be closely related or may come from the same sub-population in which the profile is more common than in the overall population. We will argue that the consequences of considering identity by descent are sufficient to include it in the calculations even if initially it seems unlikely.

1.3. The alternative hypothesis

When the first locus is typed, the observation of a match makes it more likely that the suspect is the criminal. However, several alternatives such as the 'same sub-population' or 'close relative' hypotheses are also made more likely by the match. Therefore the probability of a match at the second locus given a match at the first

locus is greater than is implied by the product rule. This effect increases with each match at subsequent loci.

We assume that the forensic DNA profile was left by the criminal. The match probability is calculated under the hypothesis I that the suspect is not the criminal. However, I must be made precise before any probability can be calculated. A natural formalization of I is that the suspect has been randomly chosen in some population of possible suspects. This population is sometimes called the 'reference population' but we will use the term 'suspect population'. In practice it is usually impossible to specify the suspect population precisely but it is routinely taken to consist of members of the suspect's broad racial group (Caucasian, Afro-Caribbean, Hispanic, etc.) since this is the population for which a database is available to provide allele frequency estimates. We will refer to the population from which the database is drawn as the 'database population'.

The usual implicit definition of I as 'random selection among unrelated individuals in the database population' is consistently unfair to innocent suspects since it ignores uncertainty about the suspect population. If the suspect is innocent then we have observed two copies of a profile which is believed to be extremely rare in the database population. However, this observation makes more likely the possibility that the profile is more common in the suspect population than in the database population. In addition, the possibility that the suspect is a close relative of the criminal cannot reasonably be excluded in general. Although it is widely recognized that match probabilities are larger for relatives, what has not previously been appreciated is that, even if this possibility is initially regarded as unlikely, the observation of matching profiles makes the 'close relative' hypothesis much more likely [19,20]. The effect is strongest in the case that the suspect is a sibling of the criminal and in practice this possibility needs to be excluded through other evidence. However, the effect can be important for other close relatives.

Much of the confusion in the debate on DNA profile analysis can be traced to the failure to address the issue of the definition of I. Weir [21] asserts that the ethnic background of the defendant (or suspect) is irrelevant. However, the argument is based on the assumption of independent suspect and perpetrator populations which is difficult to justify in general. The suspect population clearly must include the suspect, although it may also include members of ethnic groups distinct from that of the suspect. The suspect population may be specified precisely if, for example, the defence proposes a specific alternative scenario. However, such cases are rare. We argue in Section 2.1 that it will typically be plausible to assume that the suspect population consists of members of the suspect's ethnic group, or sub-population. In this case the effect of uncertainty about allele frequencies in the suspect population can be taken into account using knowledge of population genetics.

2. Genetic issues

2.1. The 'same sub-population' alternative

We propose in Section 3 a modification to the product rule in terms of a parameter F which measures our uncertainty about allele frequencies in the suspect population.

Typically, this uncertainty is due primarily to the fact that the suspect population differs from the database population.

Brookfield [15] considers the case that the suspect population consists of a number of sub-populations with mean allele frequency equal to the database value. He concludes that typically the effect of population stratification within the suspect population is small. However, we are concerned with a different problem: the fact that allele frequencies in the suspect population will in general differ from those of the database population. Reasons for apprehending innocent suspects are difficult to assess and hence the precise specification of the suspect population is usually not possible. However, an innocent suspect will typically be similar in several respects to the criminal and this similarity is connected with the reasons behind the wrong apprehension. For example, the innocent suspect may physically resemble the criminal or live in the same area. In the case of a resident of a small town accused of a crime committed locally, if the suspect is innocent it is more likely that the criminal is also from that area rather than, as Brookfield implies, that the criminal is equally likely to come from any part of the database population. Allele frequencies in the town may vary from the forensic database values. Further, due to coancestry the suspect's alleles will tend to be more common in the town than in the database population and hence ignoring the effect is consistently unfair to an innocent suspect.

We have argued that there are a priori reasons to suppose that the suspect, if innocent, is drawn from the same sub-population as the criminal. In addition, the observation of matching profiles makes this possibility more likely than before the profiles were observed, as noted in the NRC report [7, p. 79]. Finally, the 'same sub-population' assumption is conservative, since the suspect's profile will tend to be more common in his/her sub-population than in other groups. Although the suspect population will in some cases include individuals not from the suspect's sub-population, ignoring this possibility will usually be unimportant but tend slightly to favour the suspect.

Under the 'same sub-population' alternative hypothesis, the parameter F is essentially the same as the parameter known to population geneticists as F_{ST} which measures the variation in sub-population allele frequencies.

2.2. Interpretation and estimation of F_{ST}

Sewell Wright introduced F_{ST} as a measure of interpopulation variation in allele frequencies [22]. A variety of approaches to its estimation have been developed. One of the most widely used is due to Weir and Cockerham [23] who estimate F_{ST} in terms of the ratio of between sub-population variability to total genetic variability at a locus.

In simple models, F_{ST} can also be interpreted in terms of kinship: the proportion of alleles that share a common ancestor in the same sub-population and hence F_{ST} can also be estimated, for example, from the analysis of marriage records. However, interpretation of F_{ST} in terms of common ancestry can lead to some confusion. Cavalli-Sforza and Piazza [24] summarize variation among populations drawn from throughout Asia, Africa and Europe at a large number of loci. They found average F_{ST} values of 6.6%, 5.2% and 1.4%, respectively. These values are similar to the kin-

ship of cousins who share about 6.7% of their alleles through the common grandparents. It is clearly implausible for large populations to be so closely related. Instead, the high level of coancestry implied by large F_{ST} values can be explained by a long common history. As evolution has proceeded, some alleles have been lost from the population and some have spread. This will have happened by chance and because of selection. It follows that at any one locus, some of the ancestral population will have no descendants and some will have many. The longer the common history, the fewer common ancestors we expect to see. In addition, the ancient populations may have been smaller, and they may have been founded by many fewer individuals. Indeed some of the trends in human gene frequencies across Europe can be traced back to the colonization that took place during the Neolithic period [25].

The genealogies of different loci are nearly independent. If we pick an allele from each of two loci in the same individual and trace their ancestry, there is a 50% chance that it will diverge at each generation. This is simply because there is a choice of two parents. Nevertheless, the values of F_{ST} at different loci will tend to be similar because the loci will share the same history of population foundation, growth and immigration. Conversely, F_{ST} may differ among loci due to differences in the mutation process. For example, D1S7 (MS1) shows little differentiation among major ethnic groups [26], and this may be attributable to a high mutation rate which generates a stationary distribution of allele frequencies.

2.3. *Values of F_{ST}*

Extensive studies have been made of allele frequencies in many human populations to estimate values of F_{ST}. Only a few studies [27] have surveyed the VNTR loci used in forensic identification. There are some difficulties in analysing the data, but their results are broadly consistent with previous work. At conventional loci estimated values are often much less than 1% [5], and would make little difference to match probability calculations. However, the methodology and the choice of samples can be important and the appropriate studies may be those which find the larger values of F_{ST}.

The largest of the reported values may be inappropriate because they are found at loci subject to different regionally varying selection, or between sub-populations that are abnormally small, isolated and ancient. Conversely, genetic variation can be underestimated because genetically distinct populations have been pooled. For example, Devlin and Risch [27] pool data from historically distinct Texan and Californian Hispanics to form a single 'South-West Hispanic' group. Nevertheless, they illustrate genetic variation between their Hispanic groupings. The magnitude of this variation is difficult to estimate. They calculate a value equivalent to F_{ST} of 2.6%, using a method which they consider will overestimate it. However, they show how the interpretation of VNTR data is complicated by lack of resolution between alleles on gels and the large number of alleles, which means that their distribution must be described by a large number of parameters. They apply corrections that decrease variation within and between populations.

Underestimation of F_{ST} may also be a consequence of the survey covering too narrow an area to embrace distinct populations, or because recurrent mutation

generates indistinguishable alleles. For these and other reasons (see Section 3.3), we propose that it is prudent to use F_{ST} values from the large end of the currently observed range.

3. Proposed analysis

3.1. One locus: the unrelated case

Initially we will assume that the innocent suspect is drawn from the same subpopulation as the criminal but the two are not closely related. Derivations of the formulae in this section are given in Appendix A.

3.1.1. Heterozygotes

Let A and B denote the two observed alleles and write p_A and p_B for their proportions in the database population. Under I, if p_A and p_B are known, then the probability that the suspect has the alleles given that the criminal has them is

$$2\frac{(F+(1-F)p_A)(F+(1-F)p_B)}{(1+F)(1+2F)} \tag{3}$$

compared with $2p_Ap_B$ in the usual calculation. When p_A, p_B and F are all small, the match probability is approximately obtained by simply adding F to each of p_A and p_B in the usual calculation.

In practice p_A and p_B are not known exactly but are estimated from frequencies x_A and x_B in a database of size n. Taking the uncertainty due to sampling error into account is approximately achieved by adding both the suspect and the criminal profiles to the database, so that p_A and p_B are replaced with $(x_A + 2)/(n + 4)$ and $(x_B + 2)/(n + 4)$. This conclusion, which is derived in Appendix A, is based on the assumption that, before observing any data, p_A and p_B have a uniform distribution. Other assumptions are possible, but this choice is simple and conservative. Further, since the database is typically large, the match probability varies little over a range of reasonable assumptions. For a fuller discussion of the effect of sample size, see Ref. 11.

3.1.2. Apparent homozygotes

Apparent homozygotes are not necessarily homozygotes, since there may be two distinct alleles whose lengths are too similar to be distinguished. Alternatively, there may be a second 'null' allele which is not observed because, for example, it is too long or too short to appear on a gel. For convenience, we will relax the definition of homozygotes to include the former case and hence apparent homozygotes are either true homozygotes or 'null' heterozygotes.

If there were no sampling error or allele-frequency variability among subpopulations, then the match probability for apparent homozygotes would be

$$p_A(p_A + 2p_N) \tag{4}$$

in which p_A and p_N denote the population proportions of the observed and null alleles. Now p_A can be estimated in the usual way from database proportions, but p_N is difficult to estimate. For example, if no bands are observed on the gel, then it may be that there are two null alleles or there may have been some other experimental problem. It has become customary [28] to report the value $2p_A$, which is an upper bound on Eq. (4) and effectively assumes that $p_N = 1$. Null alleles are known to be rare and hence this approach is extremely conservative with respect to the value of p_N. However, other problems such as those due to population structure are ignored and therefore the value $2p_A$ may not be conservative overall.

If $p_N = 0$ and hence all apparent homozygotes are true homozygotes then, under the hypothesis of innocence, we would have observed four copies of an allele. This suggests that the allele is much more common in the suspect population than in the database population and the match probability would then be

$$\frac{(2F+(1-F)p_A)(3F+(1-F)p_A)}{(1+F)(1+2F)} \tag{5}$$

which is approximately $6F^2$ when $p_A \ll F \ll 1$. Approximate allowance for sampling variability is obtained by replacing p_A in Eq. (5) with $(x_A + 4)/(n + 4)$. In practice $p_N > 0$ and we must consider all four combinations of true homozygote and null heterozygote for the suspect and criminal alleles. The resulting match probability is

$$\frac{F+(1-F)p_A}{(1+F)(1+2F)(F+(1-F)(p_A+2p_N))} \times \tag{6}$$

$$(2F+(1-F)p_A)(3F+(1-F)p_A)+4p_N(1-F)(3F+(1-F)(p_A+p_N)).$$

An upper bound on Eq. (6) is $2(F + (1 - F)p_A)$, which is approximately the value when $p_N = 1$. This bound can be criticized as too conservative and an alternative approach is to use Eq. (6) with half the database proportion of apparent homozygotes as an upper bound on p_N.

3.2. One locus: the general case

Close relatives of the suspect cannot all be unequivocally excluded in general. Here we derive the match probability under specific relationships. A full analysis involves assigning probabilities to the possible relationships, which is discussed further in Section 4.1.

Two matching alleles could both be inherited from common recent ancestors such as parents or grandparents, otherwise they may still match with a probability that depends on the sub-population frequency. Considering each of these two possibilities at each of the two alleles the match probability becomes

$$R_2 + R_1Z_1 + R_0Z_2 \tag{7}$$

in which the relatedness coefficients R_i denote the probability that i suspect alleles

match through recent shared ancestors, $i = 0,1,2$. Values of R_i for some common close relationships are given in Table 1.

In the heterozygote case,

$$Z_1 = \frac{2F + (1 - F)(p_A + p_B)}{2(1 + F)} \tag{8}$$

and Z_2 is the match probability in the unrelated case given by Eq. (3). Similarly, in the apparent homozygote case, Z_2 is the unrelated match probability given by Eq. (6) while

$$Z_1 = \frac{(F + (1 - F)p_A)(2F + (1 - F)p_A) + p_N(1 - F)(4F + (1 - F)(3p_A + p_N))}{(1 + F)(F + (1 - F)(p_A + 2p_N))} \tag{9}$$

in which p_N is the relative frequency of null alleles.

3.3. k loci

When close relatives are accounted for, we argued in Section 2.2 that genealogies are to a close approximation independent. Further, we have restricted attention to the suspect's sub-population and hence concerns about the Wahlund effect and correlations among loci can be ignored. Therefore the whole profile match probability is, to a close approximation, the product of the single-locus probabilities.

We have assumed here that F is known, which would require knowledge of the evolutionary history of the suspect population, including details of the mutation process. Since F is not known exactly, we should allow for the uncertainty about its value by calculating the appropriate average match probability over the range of possible values of F, which would involve averaging high powers of F. The average of high powers is disproportionately affected by rare occurrences of large values. For example, a mean square is in general larger than the square of the mean. Hence a pragmatic alternative is simply to take a value of F at the high end of the plausible range. Migrant groups tend to be genetically distinct and differentiated from each other and a value of 5% may be reasonable. Even larger values may be appropriate for small isolated communities, particularly those in which consanguinous marriages are common. However for a large established population, 1–2% may suffice. Further studies involving forensic VNTR markers will allow refinement of these figures.

Table 1
Some common relatedness coefficients. The coefficients for an offspring are the same as those for a parent. The coefficients for an uncle, aunt, niece or nephew are the same as those for a half-sib.

	R_0	R_1	R_2
Sibling	1/4	1/2	1/4
Parent	0	1	0
Half-sib.	1/2	1/2	0
Cousin	3/4	1/4	0
Unrelated	1	0	0

4. Discussion

4.1. Interpretation of the match probability

Sample calculations for a four-locus, seven-allele profile are given in Table 2. The apparent homozygote match probability Eq. (6) has been used with the relative frequency of null alleles assumed to be $p_N = 0.1$. This gives a smaller single-locus match probability than the commonly used value $2p_A$ and hence the profile match probabilities given in the table are smaller in some cases than the simple product rule value. However, in the unrelated case with $F = 0.05$ and database size 100 alleles, the corrected match probability is greater than that given by the simple product rule by a factor of 38. The corrected value is so small (1.1×10^{-6}) that the practical effect of the correction may be unimportant, but the effect will be substantial in some cases.

Match probabilities are open to misrepresentation in court [29]. Consider a crime in a city of one million people in which suspect and criminal share a trait occurring in 0.1% of the population. The 'prosecutor's fallacy' is to argue that the probability of the defendant's guilt must then be 99.9%, or, in other words, the odds on guilt are 999 to 1. The 'defendant's fallacy' is to argue that 1000 people in the city share

Table 2
Match probabilities for a four locus, seven-band profile with allele relative frequencies (2%,5%), (5%,10%), (5%,15%) and (5%, —). The relative frequency of null alleles is assumed to be $p_N = 10\%$. The match probability calculated using the simple product rule is 3.0×10^{-8} (assuming $p_N = 100\%$). Database sizes (alleles): (a) 100; (b) 1000; (c) 10 000

	$F = 0.00$	$F = 0.02$	$F = 0.05$
(a)			
Unrelated	3.6×10^{-8}	1.9×10^{-7}	1.1×10^{-6}
Cousin	1.3×10^{-6}	3.8×10^{-6}	1.3×10^{-5}
Half-sib.	8.4×10^{-6}	2.0×10^{-5}	5.8×10^{-5}
Parent	7.7×10^{-5}	1.7×10^{-4}	4.0×10^{-4}
Sibling	8.4×10^{-3}	1.0×10^{-2}	1.2×10^{-2}
(b)			
Unrelated	4.9×10^{-9}	4.6×10^{-8}	4.3×10^{-7}
Cousin	4.2×10^{-7}	1.6×10^{-6}	6.6×10^{-6}
Half-sib.	3.2×10^{-6}	9.6×10^{-6}	3.3×10^{-5}
Parent	3.2×10^{-5}	8.6×10^{-5}	2.5×10^{-4}
Sibling	7.4×10^{-3}	8.6×10^{-3}	1.1×10^{-2}
(c)			
Unrelated	3.9×10^{-9}	3.9×10^{-8}	3.9×10^{-7}
Cousin	3.7×10^{-7}	1.4×10^{-6}	6.2×10^{-6}
Half-sib.	2.8×10^{-6}	8.8×10^{-6}	3.1×10^{-5}
Parent	2.9×10^{-5}	8.0×10^{-5}	2.4×10^{-4}
Sibling	7.3×10^{-3}	8.5×10^{-3}	1.1×10^{-2}

the trait and so the odds are only 1 to 999 against guilt. Clearly both arguments cannot be correct. In fact, the match probability p should be used as a likelihood ratio and interpreted together with the other evidence [30]. The odds on guilt based on the DNA match is the product of the likelihood ratio ($1/p$) and the odds based on other evidence.

Both the so called 'fallacies' are, in fact, correct if the other evidence gives the appropriate odds: evens for the prosecutor's and one to a million against for the defendant's. The latter would be obtained if other evidence pointed to all one million inhabitants of the city being equally likely suspects. Although this is rarely realistic, if there is little or no evidence other than the DNA profile, there may be a large number of 'possible suspects' and hence even a large likelihood ratio may not provide convincing proof of guilt. It is the role of the court to evaluate the other evidence, and hence to combine it with the DNA evidence. It will take some care to present likelihood ratios without causing confusion, or leading to their misuse. For a further discussion, see Balding and Donnelly [18].

Consider the case shown in Table 2b with $F = 0.05$. If other evidence does not point to the defendant's brother, it may seem reasonable to exclude him from the deliberations. In many cases this would be inappropriate. For example, assume that the other evidence can be interpreted as showing that there is an even chance that an unrelated individual commited the offence, and a 1-in-100 chance that the defendant's brother did. Hence the relative odds against guilt for the defendant, any unrelated individual and the brother are 1:1:100. In the light of the DNA evidence the brother is still the most plausible alternative to the defendant with relative odds against guilt of $100 \times 1/(1.1 \times 10^{-2}) = 9000$ compared with $1 \times 1/(4.3 \times 10^{-7}) = 2\,300\,000$ for unrelated individuals. In this case, only reporting the likelihood ratio for an unrelated individual would be misleading. If the defendant's DNA profile is the only available evidence, then the prior odds for the defendant are the same as for any of his brothers. In this case, if the defendant has b brothers then the posterior probability of his guilt is less than $1/(1 + b \times 1.1 \times 10^{-2})$, which may not be considered sufficient for a satisfactory prosecution. For example, with $b = 3$ brothers the probability of guilt is at most 0.97 here.

Statements presented in court which use a form of words similar to 'an unrelated individual is 10 000 times less likely than the defendant to have produced the DNA profile in the forensic sample' are reasonable and accurate, but it would be prudent to draw attention to two further issues. Firstly, that the likelihood ratio should be weighed together with other evidence that could increase the odds if it is corroborative, or decrease the odds if it points away from the defendant. Secondly, that unless close relatives of the defendant are excluded, they may be the most likely alternative suspects and the court should be made aware of the smaller likelihood ratio.

4.2. Further research

There is a need for further research on two fronts: estimation of true allele length distributions and studies of variation amongst human sub-populations at forensic VNTR loci. At present most information on the distribution of allele lengths in database populations is available only to the resolution imposed by the measurement

error on the gels. More precise knowledge of the distribution would help in assessing the robustness of the approach proposed by Berry et al. [31] for quantifying DNA profile evidence.

These authors propose an analysis of DNA profile evidence which dispenses with the concept of 'match' and reports an estimate of the ratio of the likelihood of all the allele length data under hypothesis I to its likelihood under G, the hypothesis of identity of suspect and criminal. This approach has important advantages: it makes more efficient use of the data and avoids the arbitrary selection of a match criterion. One major difficulty is that an estimate of the true allele length distribution is required. This poses no problem in the case of, for example, D1S7 (MS1), whose distribution appears to be smooth. However, in the case of D12S11 (MS43A) the distribution seems to be spiky and hence difficult to estimate. In addition, as the NRC report [7, p. 85] recognizes, there may be difficulties in this approach in allowing for uncertainty, such as that due to sampling error and population structure. These difficulties are probably not insurmountable and further work should be fruitful.

In general there is a need for surveys of the genetic differentiation at forensic VNTR loci among human sub-populations. As we argued in Section 2.2, the magnitude of genetic variation can differ from locus to locus, and most of our information on human populations comes from more conventional genetic markers. As the information about variation at VNTR loci accumulates, we anticipate that it will be possible to refine the values of F used in the match probability calculations. At present we have suggested large values to encapsulate the existing uncertainty.

The NRC report [7, p. 85] acknowledges that inference should properly be based on allele frequencies in the suspect population. It states that lack of knowledge about the suspect population makes the calculation impractical. We have shown that the correct calculation can be made. There are a number of sources of uncertainty about the suspect population: sampling error, how representative the data base is and population sub-division. This uncertainty can be accounted for in a reasonably straightforward and efficient way.

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Appendix A

Let p_A and p_B denote the population proportions of alleles A and B. Since initially nothing is known about the sub-population frequencies, the probability that an allele drawn in the sub-population is of type A is $P_r[A|p_A] = p_A$ and similarly $P_r[B|p_B] = p_B$. However, the probability of drawing allele A in both of two random draws from the sub-population is

$$P[A^2|p_A] = p_A(F + (1 - F)p_A \tag{10}$$

The observation of one A allele in the sub-population makes it likely that A is more common in the sub-population than in the general population and hence Eq. (10) is larger than the probability p_A^2 of drawing two consecutive A alleles in the general population. The probability of drawing first an A followed by a B allele is

$$P[AB|P_A,P_B] = p_A p_B(1 - F) \tag{11}$$

There are two approaches to justifying Eq. (10) and Eq. (11), one genetical and one statistical. The statistical argument is based on the assumption that the relative frequency of an allele in the suspect population has expectation and variance which depend only on the population relative frequency of the allele and not on the frequency of any other allele. For further details see Ref. 32, p. 59.

The genetic argument for Eq. (10) and Eq. (11) is based on assuming that $F \equiv F_{ST}$ and interpreting its value as the probability that two alleles are identical through inheritance from a common ancestor in the same sub-population. Any two alleles will be identical by descent unless there has been a mutation since their most recent common ancestor. If an ancestor of one of the alleles has entered the sub-population through migration, then it is assumed to be drawn from the same distribution as the database population. Making these assumptions, two alleles drawn from the sub-population are identical through a common ancestor in the sub-population with probability F and the ancestor is of type A with probability p_A. If not identical by descent, two alleles are of type A with probability p_A^2 and Eq. (10) follows. If the two alleles are distinct, then they cannot be identical by descent and hence Eq. (11).

If there has been a mutation since a common ancestor in the same sub-population, then the nature of the mutation process becomes an issue. VNTR loci are known to have high mutation rates, and the new alleles tend to be close in length to the progenitor allele [33,34]. If one of the of the alleles is of type A, the other could be classified as A either because its length is indistinguishable (due to the measurement error on the gels), or because there has been a more recent mutation back to type A. These possibilities increase the chances of a match being declared, but should not

lead us to underestimate match probabilities. If F has been estimated from VNTR loci, then mutations producing indistinguishable alleles will not have been scored and so the value of F will be appropriate. Alternatively, if F has been estimated from more conventional loci, then their lower mutation rates will result in slightly larger values of F, and hence the calculations will be conservative.

Both the statistical and the genetic arguments for Eq. (10) and Eq. (11) can be extended to the case of randomly drawing more than two alleles in the sub-population. In general, if $P[A^rB^s]$ denotes the probability that among $r + s$ alleles drawn randomly from the sub-population, the first r are of type A and the following s are of type B, then

$$P[A^{r+1}B^s \mid p_A,p_B] = P[A^rB^s|p_A,p_B]\,\frac{rF + p_A(1 - F)}{1 + (r + s - 1)F} \tag{12}$$

In particular,

$$P[A^4|p_A] = P[A^2|p_A]\,\frac{(2F + (1 - F)p_A)(3F + (1 - F)p_A)}{(1 + F)(1 + 2F)} \tag{13}$$

$$P[A^2B^2|p_A,p_B] = P[AB|p_A,p_B]\,\frac{(F + (1 - F)p_A)(F + (1 - F)p_B)}{(1 + F)(1 + 2F)} \tag{14}$$

The heterozygote match probability is the conditional probability that four random draws in the sub-population result in two A alleles and two B alleles, given that the first two alleles drawn are one A and one B. Hence Eq. (3) is obtained as the ratio of Eq. (14) to Eq. (11), doubled to allow for the two ways of ordering the matching A and B alleles. The homozygote match probability in the absence of null alleles is the conditional probability that four random draws in the sub-population all result in A alleles given that the first two alleles are both A and hence Eq. (5) is the ratio of Eq. (13) to Eq. (10). The apparent homozygote match probability Eq. (6) is derived similarly, allowing for all four possible combinations of true homozygote and null heterozygote in the suspect and criminal profiles. The details of the derivation of the general match probability Eq. (7) are also similar. For example, in the case of a heterozygous match in which one allele is inherited from the specified ancestor, Eq. (8) is $P[A^2B + AB^2]/2P[AB]$.

To allow for sampling error, suppose that, before observing any data, p_A and p_B are equally likely to take any two positive values which sum to less than one. Then, after noting frequencies x_A and x_B in a sample from the population of size n, we have

$$E[p_A^{r+1}p_B^s] = E[p_A^rp_B^s]\,\frac{x_A + r + 1}{n + r + s + 2} \tag{15}$$

The population proportions p_A and p_B should be replaced in Eqs. (10), (11) and (12)

by the relevant expectations given by Eq. (15). For example, we have

$$P[A^2] = \frac{x+1}{n+2}\left(F + (1-F)\frac{x+2}{n+3}\right) \tag{16}$$

instead of Eq. (10). The four-allele probabilities Eq. (13) and Eq. (14) involve expectations of powers up to four and hence the sampling correction is approximately obtained by adding all four alleles to the database.